The quantum number that describes the orientation of an orbital in space is:

A. Principle quantum number
B. Magnetic quantum number
C. Secondary(angular momentum) quantum number
D. Spin quantum number
E. None of the above

Select one:

- ○ A
- ○ B
- ○ C
- ○ D
- ○ E

B

The molar mass of compound S_2O_x = 112 g/mol

Atomic weights: S = 32, O =16 g/mol

What is the value x?

A. 3

B. 5

C. 7

D. 4

E. 6

A

Select one:

Which element has the greatest electron affinity (EA) for $_{14}Si$, $_{11}Na$, $_{18}Ar$

A. S

B. Na

C. Ar

Select one:

- A
- B
- C

Si

RAL CHEMISTRY 1-1

GENERAL CHEMISTRY 1-1 ► General ► MIDTERM EXAM

The element X (atomic number = 30) is a_____ element with _____ unpaired electrons

A. paramagnetic , 3
B. paramagnetic , 2
C. paramagnetic, 6
D. diamagnetic , 2
E. diamagnetic , 0

E

Select one:

A

Find the empirical formula for a compound that is contains 82.8% of "C" and 17.2% of "H"

(Atomic weights: C =12 g/mol, H= 1 g/mol)

A. C_2H_5
B. C_3H_8
C. C_2H_3
D. C_4H_{11}
E. C_3H_5

A

Select one:

○ A

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How many mole[s] 40g of Ca?
(Atomic weight of Ca = 40 g/mol)

A. 1.0 mole
B. 0.5 mole
C. 2.5 mole
D. 2.0 mole
E. 1.5 mole

A

Select one:
- (●) A
- ( ) B
- ( ) C
- ( ) D
- ( ) E

Calculate the mass percentage of oxygen (O) in of $Al(NO_3)_3$
(Atomic weights: Al =27 , N=14, O =16 g/mol)

A. 17.1%
B. 19.7%
C. 58.5%
D. 67.6%
E. 12.7 %

D

Select one:
- A
- B
- C

For the equation: $2C_3H_7OH + 9O_2 \rightarrow 6CO_2 + 8H_2O$

If 12 g C_3H_7OH (60 g/mol) are allowed to react with 20 g O_2 (32g/mol)

The mass of CO_2 (44g/mol) produced in grams is:

A. 13.75
B. 18.33
C. 22.91
D. 26.4
E. 17.6

B

Select one:

- A
- B
- C
- D

Which electron in the nitrogen ground state atom has the quantum number set (n, □, $m_□$, m_s) equal to (2,0,0, ½)

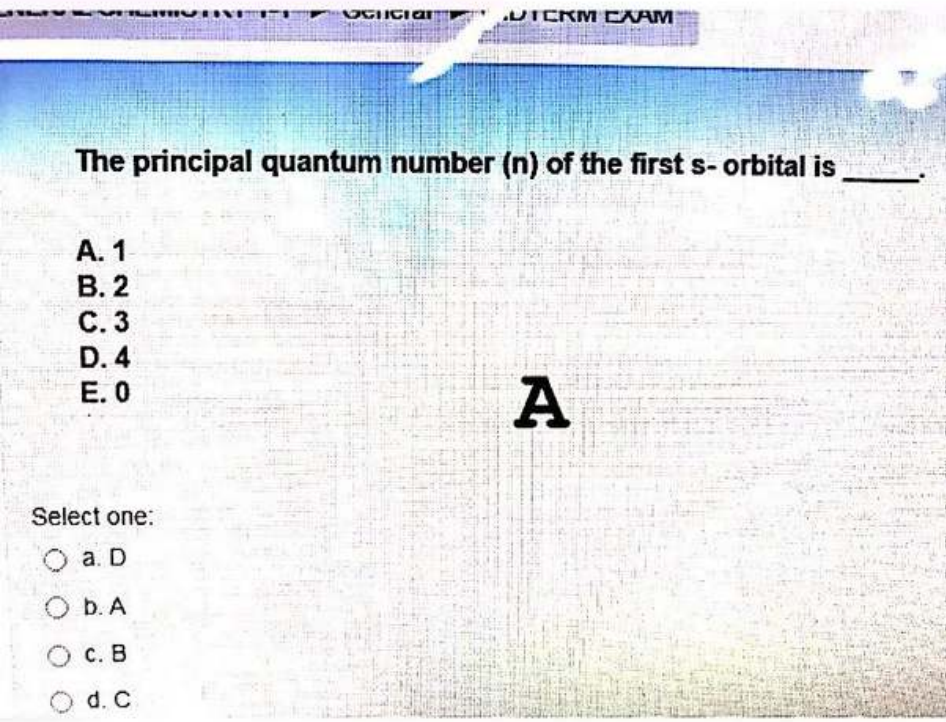
(Atomic no. of N =7)

A. first electron
B. fourth electron
C. fifth electron
D. sixth electron
E. seventh electron

D

Select one:

○ A
○ B


The principal quantum number (n) of the first s- orbital is _____.
A. 1
B. 2
C. 3
D. 4
E. 0
A
Select one:
a. D
b. A
c. B
d. C

How many electrons are there in the valence shell of the O^- ion?

(atomic number =8)

A. 2

B. 8

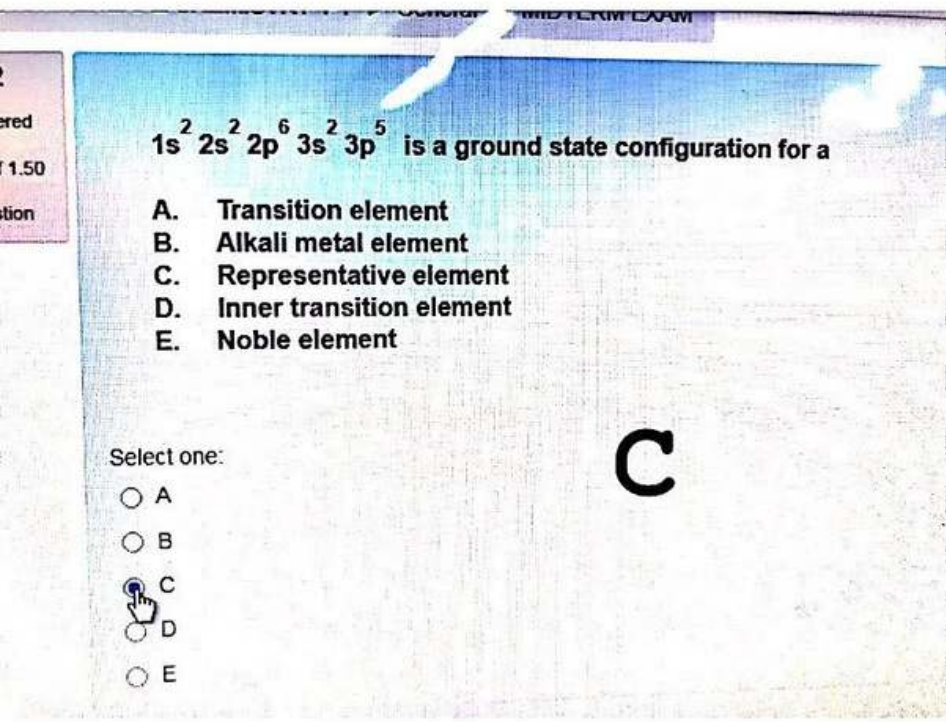
C. 7

D. 5

E. 4

Select one:

- A
- B
- C
- D


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stion
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ is a ground state configuration for a
A. Transition element
B. Alkali metal element
C. Representative element
D. Inner transition element
E. Noble element
Select one:
A
B
C
D
E
C

estion 14

yet answered

rked out of 1.50

Flag question

Which of the elements below, is a metal

A) Na (atomic no. = 11)
B) Si (atomic no. = 14)
C) B (atomic no. = 5)
D) Cl (atomic no. = 17)
E) Ar (atomic no. = 18)

A

Select one:

- A
- B
- C
- D

How many atoms (total number of atoms) are found in 7.5 mole of $K_2Cr_2O_7$

Avogadro's number = 6.022×10^{23}

A. 2.3×10^{25}

B. 7×10^{25}

C. 5×10^{25}

D. 1.4×10^{26}

E. 3.6×10^{24}

C

Select one:

Which element has the greatest IE in period 2?

A. Ne
B. Be
C. Li
D. N

A

Select one:

- A
- B
- C
- D

AL CHEMISTRY 1-1

NERAL CHEMISTRY 1-1 ► General ► MIDTERM EXAM

Aluminum reacts with oxygen to produce aluminum oxide.

$4\ Al_{(s)} + 3\ O_{2(g)} \rightarrow 2\ Al_2O_{3(s)}$

If 6.0 moles of Al react with excess O_2, how many moles of Al_2O_3 can be formed?

A. 1.5 mol
B. 2.0 mol
C. 2.5 mol
D. 5.0 mol
E. 3.0 mol

E

Select one:

- A
- B
- C
- D
- E

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01-GENERAL CHEMIS[illegible]RY 1-1

021/2020-3-30206101-GENERAL CHEMISTRY 1-1 ► General ► MIDTERM EXAM

Question 18
Not yet answered
Marked out of 1.50
Flag question

For the equation:

__ HBr + __ Al → __ $AlBr_3$ + _ H_2

The coefficients in the balanced equation (respectively) are:

A. 2,6,2,3
B. 6,2,3,2
C. 2,6,3,2
D. 6,2,2,3
E. 3,6,2,3

D

Select one
A
B

For the equation: $2\,ZnS + 3O_2 \rightarrow 2\,ZnO + 2\,SO_2$

If 3.6 mol ZnS are allowed to react with excess amount of O_2

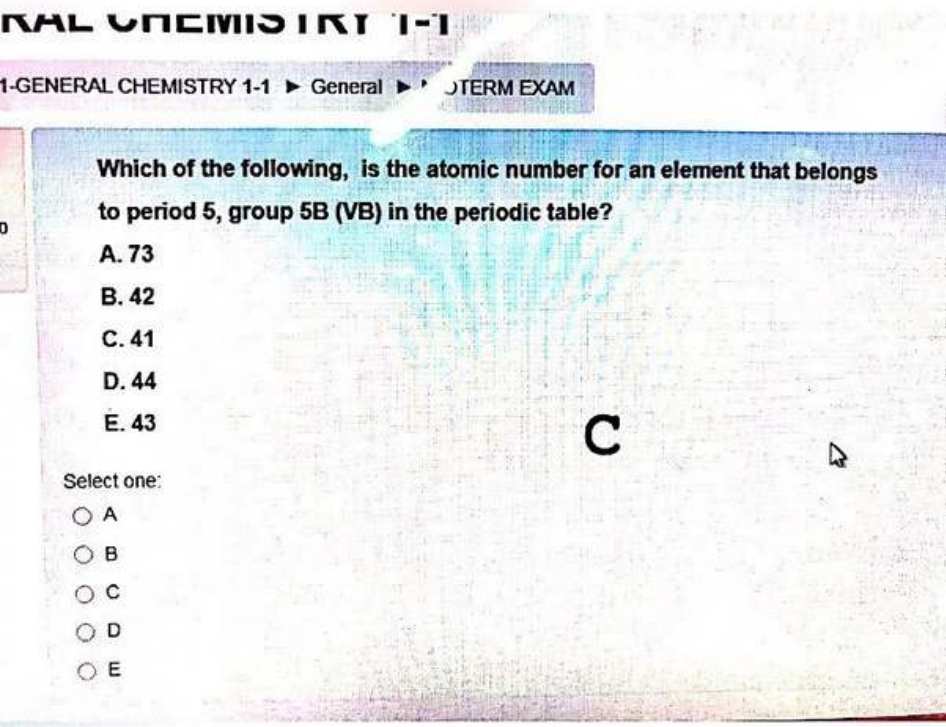
Calculate percentage yield of the reaction given that 100 g of SO_2 (64g/mol) is actually produced :

A. 86.8 %
B. 60 %
C. 43.4%
D. 26 %
E. 30 %

C

Select one:

- A
- B
- C


1-GENERAL CHEMISTRY 1-1 ► General ►
TERM EXAM
Which of the following, is the atomic number for an element that belongs to period 5, group 5B (VB) in the periodic table?
A. 73
B. 42
C. 41
D. 44
E. 43
C
Select one:
A
B
C
D
E